

1(a)(i)	<i>explanation</i> semi-conservative replication / DNA replication / S phase, has occurred ; R if replication stated to occur in, G₁/G₂ I chromatin replicates <i>difference between chromatin in G1 and G2:</i> (allow ecf from mp1 if stated replication occurs in G2) Any one from: double the / doubling in, quantity / number / amount, of, chromatin / DNA ; A chromatin has doubled sister / two identical, chromatids (formed, instead of one structure) ; A two chromatids joined at the centromere (in G2) (each) chromosome with 1 DNA molecule (in G1) has 2 DNA molecules (in G2) ; allow double, helix/ helices for molecule(s) <i>treat as neutral ref. to chromatin / chromosome, state / behaviour</i> e.g. decondensed / diffuse	2
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Question	Answer	Marks
1(a)(ii)	any three from: <i>check diagram to see if points can be awarded</i> description of nucleus as having any two listed structures ; two from: nuclear envelope, nuclear pore(s), nucleolus, nucleoplasm <i>correct spelling</i> detail of nuclear envelope ; e.g. two membranes / double membrane <i>accept from diagram</i> encloses / protects, chromatin / DNA / chromosomes / genetic information / genetic material outer membrane continuous with rough endoplasmic reticulum outer membrane with ribosomes (on outer surface) controls exit and entry of substances <i>if nuclear envelope not stated,</i> A the nucleus is surrounded by a double membrane or a double membrane encloses, chromatin / AW or (it is) double-membrane bound / bound by a double membrane detail of nuclear pore(s) ; e.g. allows (m)RNA to, pass through / leave / AW (nucleus) allows ribosomal subunits to leave (nucleus) allows entry of proteins prevents exit of DNA (small) channel / passage, through the nuclear envelope <i>can be shown on a diagram, with a label</i> detail of nucleolus ; e.g. spherical <i>can be shown as circle on diagram, labelled nucleolus</i> one / can be more than one synthesis of, rRNA / ribosomal subunits A ribosomes densely staining / dense region / condensed region <i>allow shaded in on a diagram, labelled nucleolus</i> AVP ; e.g. nucleoplasm is fluid surrounding chromatin	3

Question	Answer	Marks
1(b)(i)	R the mp if feature of cellulose stated and, is incorrect / is not neutral for mps 1, 3, 4 can accept 'cellulose does not have ..' any four from: starch has 1 (monomer of) alpha glucose / α-glucose / α-1,4 (glycosidic) bond ; I glucose / glycosidic bond / 1, 4 bond (<i>all shared with cellulose</i>) 2 monomers / glucose, orientated the same way / AW ; ora for cellulose i.e. monomers rotated 180° to each other I cellulose / cellulose molecule, rotates 180° 3 made of, two different polysaccharides / two different polymers / amylose and amylopectin ; 4 amylopectin / starch, has (α-1,4 and) α-1,6 (glycosidic) bonds ; A 1-6 if alpha noted elsewhere A link / linkage, for bonds R 'can have α-1,4 or α-1,6 bonds' 5 amylose, helical / coiled R α-helix or amylopectin branched ; <i>if both are stated, allow mp3 as well</i> A starch, is branched / has branching (v cellulose, not branched / linear) R 'starch can be helical or branched' 6 no, hydrogen / H, bonds, to hold together / between / AW, molecules / polysaccharides ; A cellulose has 7 AVP ; e.g. forms, starch, grains / granules A 'cellulose does not have ..' does not have microfibrils / fibrils / fibres A 'cellulose has..'	4

Question	Answer	Marks
1(b)(ii)	1 (different) mAbs have, different / specific, <u>antigen binding sites</u> / <u>variable regions</u> ; 1 active site 2 (each) mAb has complementary shape for, binding / AW, antigen (combining with / attaching to) or mAbs are / a mAB is, specific / complementary, to particular, antigen(s) ; 3 <i>idea that</i> a (particular) pectin / RG-I , has different, shapes / conformations / antigens, in its structure ; 4 <i>idea of</i> (using) different / specific, (sets / combinations, of) mAbs to, recognise / identify / locate / use for, different pectins / types of RG-I / a particular RG-I ; 5 <i>ref. to</i> studying development e.g ; changes occurring / side chains added, during development need different mAbs some mAbs may be specific to side chains not yet added can help to show when modifications occur 6 AVP ; e.g. reason to use more than one mAB • some mAbs may bind to more than one type of pectin • other cell wall components, may have, similar structures / same side chains • some areas of RG-I may be out of reach for a specific mAb • can, obtain information about / investigate, shape / structure of the particular pectin	3

Question	Answer	Marks
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